

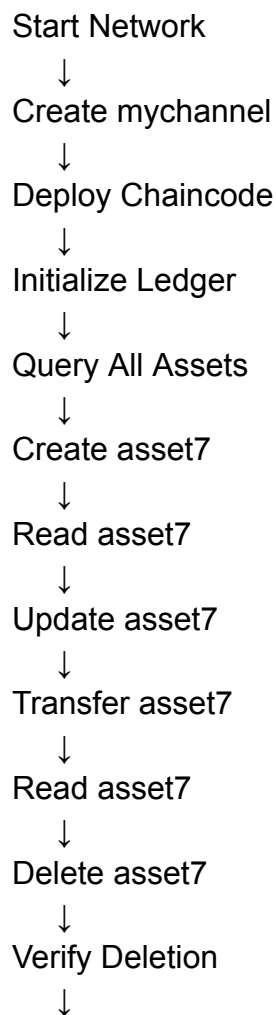
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Implementing Asset Management with Hyperledger Fabric

Aim

to implement an Asset Management application using Hyperledger Fabric by creating, reading, updating, transferring, and deleting assets through JavaScript chaincode and verifying the transactions on the blockchain ledger.

Complete flow:



Query All Assets



Stop Network

Procedure:

1. Go to the test network

None

```
cd ~/fabric-dev/fabric-samples/test-network
```

2. Start the Fabric network

None

```
./network.sh up createChannel -ca
```

3. Deploy the JavaScript chaincode

None

```
./network.sh deployCC \  
-ccl javascript \  
-ccn basic \  
-ccp ../asset-transfer-basic/chaincode-javascript
```

```
tatus (VALID) at localhost:7051  
chaincode definition committed on channel 'mychannel'  
sing organization 1  
querying chaincode definition on peer0.org1 on channel 'mychannel'...  
attempting to Query committed status on peer0.org1, Retry after 3 seconds.  
peer lifecycle chaincode querycommitted --channelID mychannel --name basic  
res=0  
ommitted chaincode definition for chaincode 'basic' on channel 'mychannel':  
ersion: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]  
query chaincode definition successful on peer0.org1 on channel 'mychannel'  
sing organization 2  
querying chaincode definition on peer0.org2 on channel 'mychannel'...  
attempting to Query committed status on peer0.org2, Retry after 3 seconds.  
peer lifecycle chaincode querycommitted --channelID mychannel --name basic  
res=0  
ommitted chaincode definition for chaincode 'basic' on channel 'mychannel':  
ersion: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]  
query chaincode definition successful on peer0.org2 on channel 'mychannel'  
chaincode initialization is not required
```

4. Set the Fabric environment

None

```
export PATH=${PWD}/../bin:$PATH
export FABRIC_CFG_PATH=${PWD}/../config/

export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_LOCALMSPID="Org1MSP"

export
CORE_PEER_TLS_ROOTCERT_FILE=${PWD}/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt

export
CORE_PEER_MSPCONFIGPATH=${PWD}/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp

export CORE_PEER_ADDRESS=localhost:7051

export
ORDERER_CA=${PWD}/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem
```

5. Initialize the ledger

Important: use both peers.

None

```
peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile "$ORDERER_CA" \
```

```

-C mychannel \
-n basic \
-c '{"function":"InitLedger","Args":[]}' \
--peerAddresses localhost:7051 \
--tlsRootCertFiles "$CORE_PEER_TLS_ROOTCERT_FILE" \
--peerAddresses localhost:9051 \
--tlsRootCertFiles
"${PWD}/organizations/peerOrganizations/org2.example.com/tlsca
/tlsca.org2.example.com-cert.pem" \
--waitForEvent

```

```

--(ram@ ram) [~/fabric-dev/fabric-samples/test-network]
$ peer chaincode invoke \
o localhost:7050 \
ordererTLSHostnameOverride orderer.example.com \
tls \
cafile "$ORDERER_CA" \
C mychannel \
n basic \
c '{"function":"CreateAsset","Args":["asset7","Black","15","Arun","900"]}' \
--peerAddresses localhost:7051 \
--tlsRootCertFiles "$CORE_PEER_TLS_ROOTCERT_FILE" \
--peerAddresses localhost:9051 \
--tlsRootCertFiles "${PWD}/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.example.com-cert.pem" \
--waitForEvent
026-09-08 17:13:36.348 IST 0002 INFO [chaincodeCmd] ClientWait → txid [7f69e66e03fc12d6e53094639f5f2eb0f7fb1f4f8386443d68917f3fcd915fb9] committed with
status (VALID) at localhost:7051
026-09-08 17:13:36.348 IST 0001 INFO [chaincodeCmd] ClientWait → txid [7f69e66e03fc12d6e53094639f5f2eb0f7fb1f4f8386443d68917f3fcd915fb9] committed with
status (VALID) at localhost:9051
026-09-08 17:13:36.348 IST 0003 INFO [chaincodeCmd] chaincodeInvokeOrQuery → Chaincode invoke successful. result: status:200 payload:"{\ID\": \"asset7\"
\color\": \"Black\", \Size\": 15, \Owner\": \"Arun\", \AppraisedValue\": 900}"

```

6. Query all assets

None

```

peer chaincode query \
-C mychannel \
-n basic \
-c '{"Args":["GetAllAssets"]}'

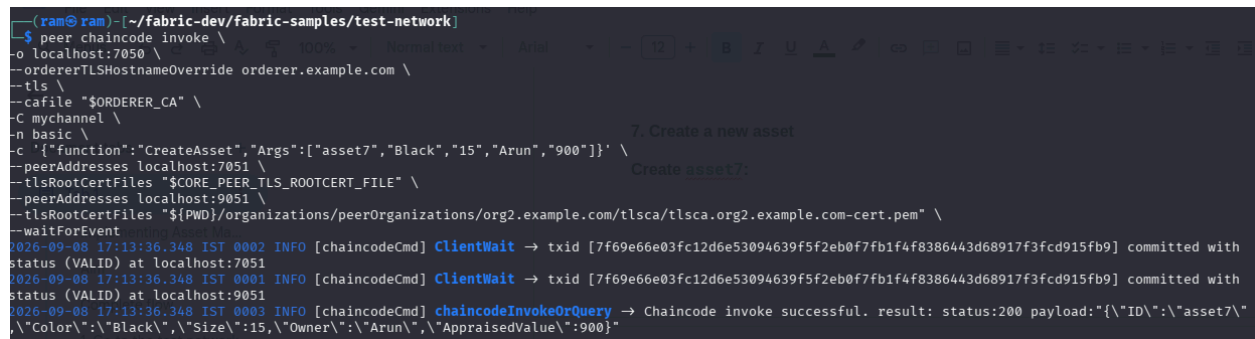
```

7. Create a new asset

Create **asset7**:

None

```
peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile "$ORDERER_CA" \  
-C mychannel \  
-n basic \  
-c '  
{ "function": "CreateAsset", "Args": [ "asset7", "Black", "15", "Arun", "900" ] }'  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles "$CORE_PEER_TLS_ROOTCERT_FILE" \  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles  
"${PWD}/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.example.com-cert.pem" \  
--waitForEvent
```



```
(ram@ram)~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile "$ORDERER_CA" \  
-C mychannel \  
-n basic \  
-c '{ "function": "CreateAsset", "Args": [ "asset7", "Black", "15", "Arun", "900" ] }' \  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles "$CORE_PEER_TLS_ROOTCERT_FILE" \  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles "${PWD}/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.example.com-cert.pem" \  
--waitForEvent  
2026-09-08 17:13:36.348 IST 0002 INFO [chaincodeCmd] ClientWait → txid [7f69e66e03fc12d6e53094639f5f2eb0f7fb1f4f8386443d68917f3fcd915fb9] committed with  
status (VALID) at localhost:7051  
2026-09-08 17:13:36.348 IST 0001 INFO [chaincodeCmd] ClientWait → txid [7f69e66e03fc12d6e53094639f5f2eb0f7fb1f4f8386443d68917f3fcd915fb9] committed with  
status (VALID) at localhost:9051  
2026-09-08 17:13:36.348 IST 0003 INFO [chaincodeCmd] chaincodeInvokeOrQuery → Chaincode invoke successful. result: status:200 payload:{"ID":"asset7",  
"Color":"Black","Size":15,"Owner":"Arun","AppraisedValue":900}
```

8. Read asset7

None

```
peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{ "Args": [ "ReadAsset", "asset7" ] }'
```

9. Update asset7

Change the value from 900 to 1000.

None

```
peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile "$ORDERER_CA" \  
-C mychannel \  
-n basic \  
-c '  
{"function": "UpdateAsset", "Args": ["asset7", "Black", "15", "Arun",  
"1000"]}' \  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles "$CORE_PEER_TLS_ROOTCERT_FILE" \  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles  
"${PWD}/organizations/peerOrganizations/org2.example.com/tlsca/  
/tlsca.org2.example.com-cert.pem" \  
--waitForEvent
```

Then verify:

None

```
peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args": ["ReadAsset", "asset7"]}'
```

```

(ram@ram)-[~/fabric-dev/fabric-samples/test-network]
$ peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile "$ORDERER_CA" \
-C mychannel \
-n basic \
-c '{"function":"UpdateAsset","Args":["asset7","Black","15","Arun","1000"]}' \
--peerAddresses localhost:7051 \
--tlsRootCertFiles "$CORE_PEER_TLS_ROOTCERT_FILE" \
--peerAddresses localhost:9051 \
--tlsRootCertFiles "${PWD}/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.example.com-cert.pem" \
--waitForEvent
2026-09-08 17:15:53.369 IST 0001 INFO [chaincodeCmd] ClientWait → txid [84d016a44ff7a2ac76ff2f9a3b86245de599e8c344440ac713552e3b55f846bb] committed with status (VALID) at localhost:7051
2026-09-08 17:15:53.370 IST 0002 INFO [chaincodeCmd] ClientWait → txid [84d016a44ff7a2ac76ff2f9a3b86245de599e8c344440ac713552e3b55f846bb] committed with status (VALID) at localhost:9051
2026-09-08 17:15:53.370 IST 0003 INFO [chaincodeCmd] chaincodeInvokeOrQuery → Chaincode invoke successful. result: status:200 payload:"{\"type\":\"Buffer\",\"data\":[]}"

```

10. Transfer ownership of asset7

Transfer from Arun to David:

None

```

peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile "$ORDERER_CA" \
-C mychannel \
-n basic \
-c '{"function":"TransferAsset","Args":["asset7","David"]}' \
--peerAddresses localhost:7051 \
--tlsRootCertFiles "$CORE_PEER_TLS_ROOTCERT_FILE" \
--peerAddresses localhost:9051 \
--tlsRootCertFiles
"${PWD}/organizations/peerOrganizations/org2.example.com/tlsca/
/tlsca.org2.example.com-cert.pem" \
--waitForEvent

```

Verify:

None

```

peer chaincode query \
-C mychannel \

```

```
-n basic \
-c '{"Args":["ReadAsset","asset7"]}'
```

```
(ram@ram)-[~/fabric-dev/fabric-samples/test-network]
$ peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile "$ORDERER_CA" \
-C mychannel \
-n basic \
-c '{"function":"TransferAsset","Args":["asset7","David"]}' \
--peerAddresses localhost:7051 \
--tlsRootCertFiles "$CORE_PEER_TLS_ROOTCERT_FILE" \
--peerAddresses localhost:9051 \
--tlsRootCertFiles "${PWD}/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.example.com-cert.pem" \
--waitForEvent
2026-09-08 17:28:28.312 IST 0001 INFO [chaincodeCmd] ClientWait → txid [f74a9906bcde4ae8ad95155f813d98edb214b603d47a99229d53f56c92960e43] committed with status (VALID) at localhost:7051
2026-09-08 17:28:28.312 IST 0002 INFO [chaincodeCmd] ClientWait → txid [f74a9906bcde4ae8ad95155f813d98edb214b603d47a99229d53f56c92960e43] committed with status (VALID) at localhost:9051
2026-09-08 17:28:28.312 IST 0003 INFO [chaincodeCmd] chaincodeInvokeOrQuery → Chaincode invoke successful. result: status:200 payload:"Arun"
```

```
(ram@ram)-[~/fabric-dev/fabric-samples/test-network]
$ peer chaincode query \
-C mychannel \
-n basic \
-c '{"Args":["ReadAsset","asset7"]}'
{"AppraisedValue":1000,"Color":"Black","ID":"asset7","Owner":"David","Size":15}
```

11. Delete asset7

Now delete the asset:

None

11. Delete asset7

Now delete the asset:

None

```
peer chaincode invoke \
-o localhost:7050 \
--ordererTLSHostnameOverride orderer.example.com \
--tls \
--cafile "$ORDERER_CA" \
-C mychannel \
-n basic \
-c '{"function":"DeleteAsset","Args":["asset7"]}' \
--peerAddresses localhost:7051 \
--tlsRootCertFiles "$CORE_PEER_TLS_ROOTCERT_FILE" \
--peerAddresses localhost:9051 \
```

```
--tlsRootCertFiles
```

```
"${PWD}/organizations/peerOrganizations/org2.example.com/tlsca  
/tlsca.org2.example.com-cert.pem" \
```

```
--waitForEvent
```

```
(ram@ram)-[~/fabric-dev/fabric-samples/test-network]  
$ peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile "$ORDERER_CA" \  
-C mychannel \  
-n basic \  
-c '{"function":"DeleteAsset","Args":["asset7"]}' \  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles "$CORE_PEER_TLS_ROOTCERT_FILE" \  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles "${PWD}/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.example.com-cert.pem" \  
--waitForEvent  
2026-09-08 17:29:46.080 IST 0001 INFO [chaincodeCmd] ClientWait → txid [1f251d25562066ba4bb3ea5c97900fdab4f89f3ab15b46d74106f66221aad387] committed with  
status (VALID) at localhost:7051  
2026-09-08 17:29:46.080 IST 0002 INFO [chaincodeCmd] ClientWait → txid [1f251d25562066ba4bb3ea5c97900fdab4f89f3ab15b46d74106f66221aad387] committed with  
status (VALID) at localhost:9051  
2026-09-08 17:29:46.081 IST 0003 INFO [chaincodeCmd] chaincodeInvokeOrQuery → Chaincode invoke successful. result: status:200 payload:{"type":"Buffer  
","data":[]}"
```

12. Verify asset7 was deleted

None

```
peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["ReadAsset","asset7"]}'
```

You should get an error indicating that the asset doesn't exist.

```
(ram@ram)-[~/fabric-dev/fabric-samples/test-network]  
$ peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["ReadAsset","asset7"]}'  
Error: endorsement failure during query. response: status:500 message:"The asset asset7 does not exist"
```

13. Query all assets again

None

```
peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["ReadAsset","asset7"]}'
```

```
-c '{"Args":["GetAllAssets"]}'
```

asset7 should no longer appear.

```
(ram@ram) ~/fabric-dev/fabric-samples/test-network
$ peer chaincode query \
-C mychannel \
-n basic \
-c '{"Args":["GetAllAssets"]}'
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"}, {"AppraisedValue":400,"Color":"red","ID":"asset2","Owner":
:Brad","Size":5,"docType":"asset"}, {"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":10,"docType":"asset"}, {"AppraisedValue":6
00,"Color":"yellow","ID":"asset4","Owner":"Max","Size":10,"docType":"asset"}, {"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":
15,"docType":"asset"}, {"AppraisedValue":800,"Color":"white","ID":"asset6","Owner":"Michel","Size":15,"docType":"asset"}]
```

14. Stop the Fabric network